



Chapter 11

Measuring Gene Expression via mRNA Abundance in *Candida auris*

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Abstract

The recently emerged human pathogenic yeast *Candida auris* has become a major global threat. As compared to other *Candida* species, *C. auris* often displays a high level of resistance to commonly used antifungals and poses additional therapeutic challenges. There is a great need to understand the molecular basis of its success as a drug-resistant human pathogen. The study of condition-specific gene expression can provide good cues of regulatory circuitry governing high drug resistance. Here, we describe the protocol of quantitative reverse transcription PCR (RT-qPCR) which can be conveniently employed as a highly reproducible method for measuring individual transcripts in *C. auris* cells.

Key words *Candida auris*, Drug resistance, RNA, Polymerase chain reaction, Transcriptome, Quantitative reverse transcription

1 Introduction

Gene expression in *C. auris* is highly responsive to various environmental conditions which aids in its rapid adaptation to diverse niches within the host [1, 2]. A variety of in vivo and in vitro techniques have been used to measure gene expression. Among the various methods used for expression studies, quantitative reverse transcription PCR (RT-qPCR) represents a sensitive, reliable, and highly reproducible method for measuring individual transcripts [3]. RT-qPCR meets the need for molecular medicine, biotechnology, microbiology, and diagnostics for quantitative data analysis and has become the tool of choice for mRNA quantification.

In this chapter, we provide the detailed procedure from the isolation of total RNA from *C. auris* cells, complementary DNA (cDNA) preparation, and the setup of RT-qPCR reactions. As an example, we also show relative condition-specific (antifungal treat-

ted vs untreated samples) quantification of transcripts to calculate the fold differences of mRNA levels. Common useful tips and hints are also included to help beginners understanding the technique.

2 Materials

Reagents for RNA and cDNA sample preparation should be made in diethyl pyrocarbonate (DEPC)-treated ultrapure DNase-/RNase-free water. The workbench should be kept clean with RNaseZap.

2.1 Growth

1. Yeast extract peptone dextrose (YEPD) medium: 2% Bacto Peptone, 1% yeast extract, and 2% dextrose in deionized water for broth. YEPD agar plates are prepared by adding 2% agar in the YEPD broth recipe. The broth needs to be autoclaved before it can be used.
2. 50-mL falcon tubes and 200-mL conical flasks.

2.2 RNA Isolation and cDNA Preparation

1. Microcentrifuge tubes.
2. RNaseZAP (RNase decontamination solution).
3. DEPC-treated water.
4. Acid-washed glass beads (0.4–0.6 mm).
5. DNase I 1 U/ μ L.
6. 10 $\times$ reaction buffer with MgCl₂ (DNase I treatment).
7. 50-mM EDTA: 18.61-g EDTA disodium salt dissolved in 1-L deionized water and adjusted to the pH 8.0.
8. RevertAid First Strand cDNA Synthesis Kit.
9. Agarose.
10. Ethidium bromide.
11. 1 $\times$ TAE buffer: 40-mM Tris, 20-mM acetic acid, and 1-mM EDTA.
12. Taq polymerase.
13. Standard Taq polymerase reaction buffer.
14. dNTPs.
15. RNA isolation kit.
16. Electrophoresis gel apparatus.
17. DNA marker/DNA ladder.
18. NanoDrop™ Spectrophotometer.
19. UV–Vis Spectrophotometer.
20. Yeast–Cell Disruptor.

2.3 RT-qPCR

1. Oligonucleotide primers.
2. SYBR Green Supermix.
3. 96-well plates.
4. Real-time PCR machine.

3 Methods

3.1 Cell Growth

1. For the primary culture, inoculate a single colony of *C. auris* from a YEPD agar plate in 10-mL YEPD broth, and grow for 24 h at 30 °C in a shaking incubator at 200 rpm.
2. Inoculate O.D.₆₀₀ of 0.2 in 50 mL of YEPD media for secondary culture. Grow the cells for another 7 h until they reach log phase, and use these cells for the control sample (*see Note 1*).
3. For samples with drug treatment, after 6 h of secondary growth, add the nonlethal concentrations of the desired drug, followed by growth for another 1 h.

3.2 RNA Isolation and Quantification

1. After 7 h of growth, harvest the cells by centrifugation at $3200 \times g$ for 5 min at 4 °C. Wash the cell pellet twice with DEPC-treated water (*see Note 2*).
2. Measure the O.D.₆₀₀ cells after washing. Use around 5×10^6 – 1×10^7 cells for the disruption step.
3. RNA isolation is carried out by using an RNA isolation kit as per the manufacturer's protocol (*see Note 3*).
4. The first step includes lysis of the cells (5×10^6 – 1×10^7 cells) by the mechanical disruption method. Add 300 µL of acid-washed 0.4- to 0.6-mm glass beads to the cell pellet along with 600 µL of lysis buffer provided with the RNA isolation kit.
5. 3 rounds, each of 20 s pulse are used for cell disruption (*see Note 4*). Incubate the cells on ice for 1 min after each pulse to keep the samples cold.
6. After disruption, beads and cell debris are separated by centrifugation for 5 min at $18,000 \times g$. The upper layer of cell lysate is used for RNA preparation as per the manufacturer's protocol.
7. Finally, the total RNA is eluted in RNase-free water.
8. Check the RNA concentration and purity on a NanoDrop™ 2000 Spectrophotometer or any similar measuring device. Different dilutions of RNA such as 1:10 or 1:50 should be prepared in DEPC water to determine the concentration.
9. The spectrophotometric absorption ratio between 260 and 280 nm (A_{260}/A_{280}) is a reliable measure to assess the purity of the extracted RNA. An absorption ratio between 1.8 and 2.0 (a value of A_{260}/A_{280} ratio closer to 2 is best) is an indicator of good quality of the prepared RNA (*see Note 5*).

3.3 cDNA Synthesis

1. To start with the cDNA synthesis, the first crucial step is to remove DNA contamination. For this, treat the RNA samples with DNase I using the following reaction mixture 6.25 μL of RNA (1.6- $\mu\text{g}/\mu\text{L}$ stock solution), 1 μL of 10 $\times$ reaction buffer, and 1- μL (1 U/ μL) DNase I enzyme. Make up the final volume to 10 μL by adding the nuclease-free water for each reaction. Incubate the reaction mixture at 37 °C for 30 min in a thermocycler [4].
2. Add 1 μL of 50-mM EDTA into the reaction mixture, and incubate at 65 °C for 10 min to stop the DNase treatment reaction. This DNase I-treated RNA sample is used as a template for the downstream cDNA synthesis process (*see Note 6*).
3. For cDNA synthesis use the RevertAid First Strand cDNA Synthesis Kit, and set the reaction with the following components: 1 μg of DNase I treated RNA, 1 μL of the random hexamer and/or 1 μL oligo(dT)₁₈, 4- μL reaction buffer from 5 $\times$ stock, 1 μL of RiboLock RNase inhibitor, 2 μL of 10-mM dNTP mix, and 1 μL of RevertAid M-MuLV RT enzyme. Adjust the volume of the reaction mixture is to 20 μL with the nuclease-free water (*see Note 7*).
4. For oligo(dT)₁₈-primed cDNA synthesis, incubate the samples at 42 °C for 60 min. For random hexamer primed synthesis, incubate samples for 10 min at 25 °C followed by 60 min incubation at 42 °C. Terminate the reaction by heating at 70 °C for 10 min in both cases (*see Notes 7 and 8*).

3.4 RT-qPCR and Expression Analysis

3.4.1 Primer Design and Optimization

Primer design is a crucial step for the success of the RT-qPCR assay. The quality and reliability of outcome are significantly dependent upon RT-qPCR primers that otherwise would poorly anneal or bind nonspecifically during amplification. There are many tools available to design the primers such as the NCBI primer blast tool (<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>) or Primer3. We used the Primer3 web-based tool (<https://primer3.ut.ee/>) for designing the primers on the default setting (for details *see Notes 9, 10, and 11*).

1. To validate each primer pair for specific size and specific amplification of the product, a normal PCR reaction with genomic DNA as a template and the gene specific primers should be performed. The PCR product should be analyzed via agarose gel electrophoresis to confirm the specific product at the accurate size along with appropriate DNA size marker.
2. The primer's efficiency for equal amplification should be optimized with a different dilution of the genomic DNA as a template. In an earlier study of *C. auris* gene expression analysis, we have used genomic DNA template in dilutions ranging

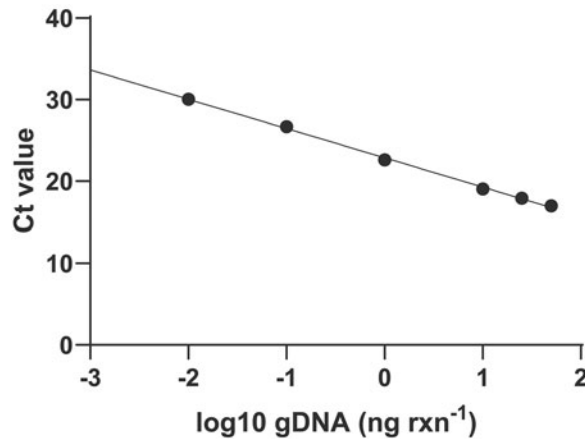


Fig. 1 Primer efficiency curve generated using the Ct values obtained with different dilutions (ranging from 0.01 ng to 50 ng) of genomic DNA as template. The calculated slope value of the plot is -3.456 , based on data adapted for *TDH1* gene primers from an earlier study [2]

between 0.01 ng and 50 ng (0.01, 0.1, 1.0, 10, 25, and 50 ng) for setting up the PCR reactions [2].

3. The standard curves are generated by plotting the graph between the $\log_{10}$ values of DNA dilutions against the Ct value as shown in the Fig. 1. A linear relationship (indicated by a straight line) shows that gene primer pairs have equal amplification for a given amount of template. The efficiency of the PCR should be between 90% and 100% having slope values between -3.3 and -3.6 .

3.4.2 RT-qPCR Reaction Setup and Selection of Endogenous Controls

To determine the relative expression of different genes, the expression of the target gene of interest must be compared to a housekeeping gene as a control. The expression of the selected housekeeping gene should not be variable under any experimental conditions (*see* **Note 12**). The endogenous control gene not only serves as a control for quality (integrity and to some degree purity) of the RNA in the reaction but also works as a control for the RT step of the RT-PCR quantification.

1. We have used *C. auris* *TDH1* (*CAUR_02457*), an ortholog of the *Saccharomyces cerevisiae* housekeeping gene *TDH1* as an internal control for normalization of expression.
2. The expression of *TDH1* (*CAUR_02457*) does not change significantly in all the tested conditions (drug treatment) justifying it as a good candidate for internal control.
3. The PCR reaction volume size can vary, but a typical PCR reaction is set up in a 25- μ L final volume for the analysis. The

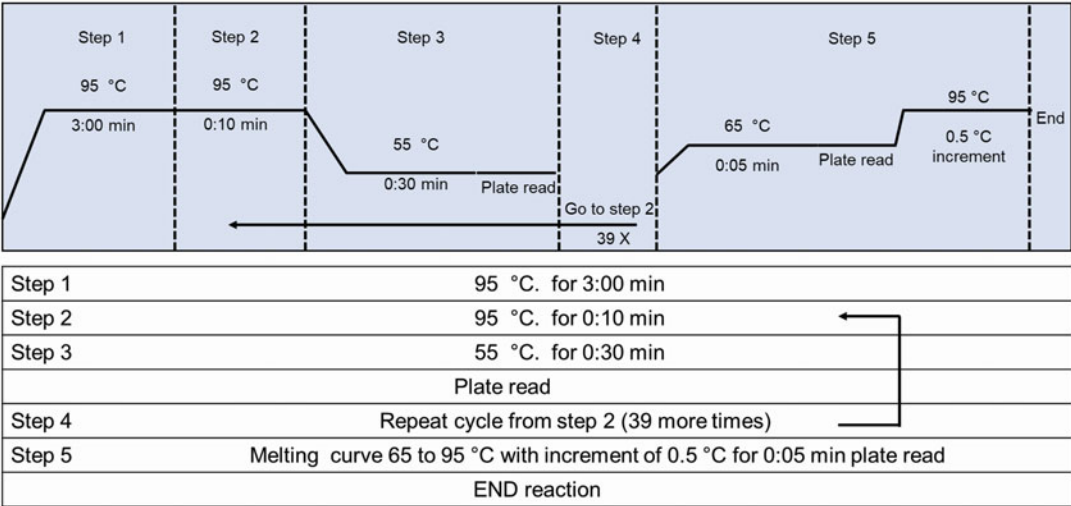


Fig. 2 Cycling conditions used for RT–qPCR reactions on LightCycler™ real-time PCR machine

qPCR should be performed in three technical replicates for each sample (*see* **Note 13**).

4. The reaction is set up by using cDNA with SYBR Green Super-mix according to the manufacturer’s instructions in a real-time PCR machine for 40 cycles in the total volume of 25 µL. The each PCR reaction consists of 12.5 µL of 2× SYBR green super mix, 0.5 µL of each primer (stock concentraion10 µM), and 1 µL of cDNA (1/10 dilution of cDNA obtained after *see* Subheading 3.3) and adjusted with the nuclease-free water to make up the volume to 25 µL. Typical PCR reaction cycling conditions are shown in Fig. 2.

3.4.3 Expression Profile Analysis

The use of fluorescent reporter dyes in RT–qPCR allows us to combine the amplification and detection steps of the PCR reaction in a single reaction [5, 6]. RT–qPCR is based on measuring the variation in fluorescent signal as a function of the amount of DNA produced during each PCR cycle. Each PCR reaction consists of the individual cycles and the number of cycles required to cross the threshold fluorescent signal called threshold cycle (Ct) or crossing point (Cp). The Ct value has an inverse relationship with the amount of starting genetic material; the lower the Ct value, the greater the amount of starting genetic material. The method used to calculate the relative expression changes of two different samples/condition is the comparative Ct method or $\Delta\Delta C_t$ method [7]. The comparative Ct method has the advantage of eliminating the need for a standard curve. However, before employing the comparative Ct method, a validation experiment must be carried out to demonstrate that the efficiencies of the target and

Table 1
Table showing the calculation of relative expression of *CAUR_01276* measured by the $2^{-\Delta\Delta Ct}$ method in the presence of amphotericin B (0.7 µg/mL) and data calculated as mean ± S.E.

Ct of <i>TDH1</i> (Untreated)	Ct of <i>TDH1</i> (treated)	Ct of <i>CAUR_01276</i> (untreated)	Ct of <i>CAUR_01276</i> (treated)	ΔCt <i>TDH1</i> (untreated)	ΔCt <i>CAUR_01276</i> (treated)	$\Delta\Delta Ct$ (ΔCt treated- ΔCt untreated)	Relative fold-change $2^{-\Delta\Delta Ct}$
14.212	14.523	23.335	21.598	9.123	7.075	-2.048	4.135
14.182	14.601	23.343	21.622	9.161	7.021	-2.140	4.407
14.127	14.555	23.304	21.643	9.177	7.088	-2.089	4.254
Mean ± S.E.							
14.174 ± 0.025	14.559 ± 0.023	23.327 ± 0.012	21.621 ± 0.013	9.154 ± 0.016	7.062 ± 0.020	-2.092 ± 0.027	4.266 ± 0.079

Adapted from a previous study [2]

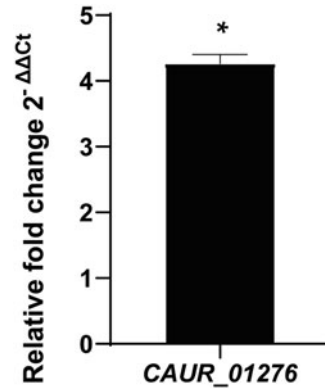


Fig. 3 Drug-induced expression profiles of *CAUR_01276* in *C. auris* checked by RT-qPCR. Data was measured by the $2^{-\Delta\Delta C_t}$ method in the presence of amphotericin B (0.7 $\mu\text{g/mL}$). Data is presented as mean $\pm$ S.E. (error bars) from technical triplicates ($n = 3$)

endogenous control amplifications are nearly equal (as described in *see* Subheading 3.4.1).

The first step is to calculate the C_t value by subtracting the C_t value of the internal control (the example here is *TDH1*) from the C_t value of the gene of interest (e.g., *CAUR_01276*) in the treated (e.g., amphotericin B treatment) and untreated conditions, which gives the ΔC_t values for the treated and untreated conditions, respectively. The next step is to find the $\Delta\Delta C_t$ value by subtracting the ΔC_t of the untreated sample from the treated one and finally putting the value of $\Delta\Delta C_t$ in the formula $2^{-\Delta\Delta C_t}$ to get the relative fold-change in the expression of the gene (*CAUR_01276*) (Table 1 and Fig. 3). Formulas and details of calculations and steps are given below.

Step 1: ΔC_t calculation for treated and untreated samples.

$$\Delta C_t \text{ treated} = C_t (\text{CAUR}_{01276}) - C_t (TDH1) \text{ (amphotericin B-treated sample)}$$

$$\Delta C_t \text{ untreated} = C_t (\text{CAUR}_{01276}) - C_t (TDH1) \text{ (no drug treatment sample)}$$

Step 2: $\Delta\Delta C_t$ calculation

$$\Delta\Delta C_t = \Delta C_t \text{ treated} - \Delta C_t \text{ untreated}$$

Step 3: Relative fold-change/ $2^{-\Delta\Delta C_t}$ calculation

Now relative fold-change can be obtained by 2 to the power of the negative $\Delta\Delta C_t$ relative fold-change upon treatment = $2^{-\Delta\Delta C_t}$

4 Notes

1. The total growth time for the secondary culture is 7 h at 30 °C, in which the culture of *C. auris* CBS10913T strain reaches the mid-log phase. The time of secondary culture growth and drug treatment can be decided as per specific experimental requirements and the strain type.
2. It is recommended to prepare a fresh culture for RNA. However, the cell pellet can also be stored at –80 °C and can be used for up to 6–9 months for RNA preparation. RNA should be isolated in triplicates from three independently grown biological samples.
3. RNA isolation can be done via a manual method using TRIzol (available under different names from different manufacturers) or via a kit-based method. We use an isolation kit (RNeasy Mini Kit, Qiagen) because it is simple to use, saves time, and produces high-quality RNA for downstream applications. The kit includes the lysis solution, wash solution, elution buffer, and mini spin columns, as well as collection tubes and elution tubes.
4. Mechanical disruption with glass beads is a low-cost and effective method of yeast cell lysis. For efficient cell lysis, we use 2-mL microcentrifuge tube with flat bottom as the lysis is more efficient as compared to conical tubes. There are many disruptors available in market; we have used Fast Prep-24A Classic cell disrupter, MP Biomedicals for cell disruption. It is recommended to optimize the cycle time as per specific disruptor used.
5. The RNA isolated can be stored at –80 °C and used in the future after a quality check.
6. After DNase I treatment, the success of the reaction can be checked by setting a PCR reaction for amplification of a housekeeping gene using DNase I-treated and DNase-untreated sample as templates. The untreated sample should provide the amplification product which can be used as a PCR reaction control, while the DNase I treated sample should not produce any amplification product.
7. The targets of interest and primer location should be considered when deciding between oligo(dT) and random primers. Using only the random hexamer is not ideal for some RT PCR reaction because of over estimation of mRNA copy number [8]. Similarly, the use of only oligo(dT) also leads to 3' bias in some targets. In two-step RT–PCR, a mixture of oligo(dT) and random primers is frequently used to maximize the benefits of each primer type. We have used both oligo(dT) and random hexamer primers together in the same reaction for cDNA

synthesis with reaction setup for incubation at 25 °C for 10 min, followed by another 60 min incubation at 42 °C. The reaction is then terminated by heating the reaction mixture at 70 °C for 10 min.

8. cDNA can be aliquots in tubes and can be stored at −20 °C up to 1 month for further use.
9. Choose primers that will provide a uniform PCR product size of 150–200 bp having a T_m of 55–60 °C for annealing, a GC content of around 40–60% to ensure maximum product stability, and the pair with the least self-complementarity.
10. The region chosen for transcription analysis is normally the mid-region of the gene of interest.
11. To avoid amplification of intronic regions, blast these primers against each gene sequence using the NCBI tool primer blast, and those with binding sites only in the exonic region are chosen for RT–qPCR assay.
12. The selection of internal control housekeeping genes should be evaluated before subjecting samples for RT–qPCR. In the case of *C. auris*, under some conditions, a variation in the expression of housekeeping genes can be noticed.
13. RNA isolation should be performed with at least two biological replicates, and RT–qPCR should be performed in experimental triplicates.

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